Electricity Demand Forecasting: A Review of Statistical, Machine learning, Deep Learning and Hybrid Approaches

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***Abstract*—Electricity demand forecasting is highly important for the planning, operation, and stability of electricity grid and power systems. The growing complexity of load patterns—driven by renewable integration, urbanization, and climate variability—has led shift from the conventional methodologies for the demand prediction such as the statistical and probabilistic approaches towards the more advanced machine learning (ML) and deep learning (DL) approaches. This review purpose focuses on the insights from recent literature covering statistical, ML/DL, and hybrid forecasting models, with particular attention to accuracy, adaptability, practical applications and their limitations. The comparative findings highlight that while statistical methods remain effective for linear and short-term trends, ML and hybrid models provide superior accuracy in large & complex, nonlinear, high-dimensional, and seasonal data and context. Challenges such as data quality, interpretability, explainability, reasonability and computational costs are discussed, alongside future opportunities for integrating AI with smart grids and renewable-centric systems.**

Keywords— electricity demand forecasting, electricity load forecasting deep learning, hybrid models, statistical methods, forecasting based on time-horizons.

I. Introduction

Electricity Demand forecasting has a Pivotal Role in architecture and functioning of the modern day power System by providing accurate predictions for the future electricity load and demand over the grid [1], which are necessary for effective operational planning, resource management and data driven decision making [1]. An Accurate forecasting of electricity demand and load is very important since an inaccurate forecasting may result in resource and economic losses [1]. Even subtle nuances in the electricity load forecasting can heavily impact the economic resources of the grid, overestimation can lead to economic wastage while underestimation can lead to deterioration and grid instability [1]. Electricity demand forecasting is majorly classified on the basis of time horizons [7]. The classification is presented as follows:

- short Term forecasting: Time frame of this forecasting ranges From couple of Minute and can extend upto One Day or even a full upcoming week [7].
    - Applications: for scheduling, real-time dispatch, market bidding [7], and security analysis [3], [7]
- medium Term forecasting: Time frame for medium frame ranges from a few day or week and extends upto full 1 year [7], [9]. While sometimes it can also be used to describe forecasting of demand on a monthly basis.
    - Applications: for maintenance, planning, load dispatching and energy trading [5], [7].
- Long Term Forecasting (LTF): The time frame for this forecast horizon ranges from two-three years and can extend up to a period of ten years with monthly granularity [7]. It involves predictions for extended period
    - Applications: for infrastructure investment, strategic system planning, capacity expansion, and policy planning and data driven long-term decision making [1], [5], [7].

The electricity demand forecasting"meth'dology has observed a shift in approaches over the time [1], [8]. Historically, linear statistical methods (for example ARIMA – Autoregressive Integrated Moving Averages) have been widely adopted since they provide transparency and robustness [4]. However, demand is increasingly influenced by exogenous factors including weather, socioeconomics, renewable variability and consumer behaviour [6]. These nonlinearities often reduce effectiveness of linear models. The integration of smart meters and IoT devices has also increased the volume of granularity of the available data, creating opportunities for AI-based models [2].

II. Statistical Approaches

Initially, electricity demand was highly predictable and forecasted using simple trend extrapolation methods, such as the "7% rule," which makes the assumption that the energy consumption get double in a period of ten years [1]. This was followed by the adoption of classical statistical models, including Moving Averages (MA) model, Auto-Regressive (AR) model [1], and later on by more complex time-series statistical models such as Autoregressive Moving Average (ARMA), Autoregressive Integrated Moving Average (ARIMA) [1], [2], [10], SARIMA [13] and regression analysis which have been widely used for time-series electricity forecasting [1]. Even though the statistical model fits well on the data showcasing linear pattern, they struggle significantly with the nonlinear variations in the electricity load pattern. For instance, the m.a.p.e of arima varies between 3–4% in several studies [2]. In case of the Regression models focused on incorporating the macroeconomic variables such as the Gross Domestic Product (GDP) and industrial output have also been proven effective for long-term electricity load forecasting [7]. But as stated earlier these statistical models takes the assumption of linearity and stationarity in the data, this makes such models struggle in handling complex and non-linear relationship which is quite prevalent in the contemporary electricity data, thereby limiting their scalability in dynamic environments [3], [10].

III. Machine Learning And Deep Learning Approches

Studies observed that ml and dl methods perform way more efficiently in handling more complex nonlinear and high-dimensional data in comparison to the conventional Statistical methods [1]. This was boosted with the introduction of smart meters which resulted in providing more granular and accurate data [2]. Popular machine Iearning models such ANN, Svr, svm, and decision Trees showcased better performance and accuracy on complex nonlinear data when compared with the traditional statistical approaches [2], [9], [13]. The next significant shift in the forecasting of electricity load and demand was towards Deep learning, which is a subset of Machine learning, based on the Neural Network Architecture with a large number of hidden layers in the architecture [3]. These models were capable of capturing more nuanced features and patterns from the data [3]. The Sequential model architectures such as Recurrent Neural Networks (RNNs), Gated Recurrent Units (which is an updated and more advanced version of RNNs) and Long Short-Term Memory (LSTM) are specifically suited for sequential time-series forecasting [4], [5]. CNN-LSTM hybrids have further improved performance by capturing spatial-temporal correlations. For example, deep regression models achieved an R^2 of 0.93 and MAPE of 2.9% in short-term load forecasting [3]. Transformer's self-attention mechanism also provides them increased ability to capture long-range dependencies in the date. Despite their accuracy, these models require substantial computational resources and a huge corpus of labelled datasets, and often act as 'black-boxes,' reducing interpretability [1], [9], [13].

IV. Hybrid Approaches

The current trend is the increasing use of hybrid models [1][8]. Hybrid forecasting integrates the interpretability of statistical models with the adaptability of ML/DL to achieve higher accuracy and evidently overcomes the limitation of any single method. [1], [13]. Examples include LSTM-XGBoost [11], ARIMA-LSTM, regression-ANN, and ANFIS-based hybrids [5].

- LSTM-arima hybrids: arima captures the Linear Trends while lstm models non-linear residuals. These approaches consistently reduce error rates compared to either method considered individually [5], [10].
- ANFIS: It fuses Fuzzy Logic with NN, enabling both interpretability and learning ability. Reported to outperform standalone ANN or LSTM [6].
- Regression + Neural Networks: Effective in long-term forecasts (LTF) where socioeconomic indicators interact with unstable factors such as weather [5]
- XGBoost + BiLSTM: This model combined the benefits of LSTM (Long Short-Term Memory) specifically Bi-directional LSTM, which is powerful for learning temporal sequential dependencies in load data, but struggles with forecasting peak loads with the advanced machine learning algorithm XGBoost (Extreme Gradient Boosting) which can handle nonlinear patterns and sharp spikes better [11].
- Prophet-LSTM: In the hybrid approach the Prophet model is used to predict the electricity load data by handling the linear & non-linear components, but it leaves some non-linear residuals [12]. These residuals (error terms from Prophet) are then trained using LSTM which handles sequential dependencies and learns the hidden non-linear fluctuations in load data [12]. The output forecast of Prophet (mostly linear part) + LSTM forecast (handling non-linear residuals) are combined using a Backpropagation Neural Network (BPNN) which acts as a lightweight optimizer reducing overfitting and improving overall accuracy by adjusting the weights [12].

These hybrid models reduce error residuals of purely statistical approaches while leveraging ML's strength in handling nonlinearities. A case study in Ukraine demonstrated that combining macroeconomic regression with LSTM achieved ~96.8% accuracy in long-term hourly forecasts [5]. Seasonal hybrid models have also been shown to adapt well across varying demand patterns and weather conditions [5]. Recent reviews emphasize hybrid models as the most promising direction, balancing flexibility, interpretability, and predictive accuracy [1].

V. PROPOSED Approach

We propose our approach to adopt and implement the methodology of the XGBoost-LSTM hybrid model. While the XGBoost-LSTM hybrid approach has been previously explored [2], we propose to apply for the purpose of electricity load data from the states of India, particularly the datasets available at the "AIKosh", an open source dataset and model online repository, we also propose to enhance the contemporary approach by introducing and integrating Explainable AI functionality in our proposed approach to tackle challenge of the AI based and hybrid models acting as

"Black Boxes". Our proposed approach tends to use SHAP/ LIME to add explainability for the XGBoost part. In addition to this we also propose to test and apply other approaches in the LSTM part of the method, we propose to use Attention based LSTM and Seq2Seq model with attention.

The fundamental approach remains the same the model aim to combine benefits from LSTM which shows high performance capability in learning temporal sequential dependencies in load data, with the advanced machine learning (deep learning) algorithm XGBoost (Extreme Gradient Boosting) which can handle non-linear patterns and sharp fluctuations more efficiently [2].

The proposed approach to introduce explainability not only provides a much enhanced and reliable approach for interpreting the reasoning behind the decision making by the model but also aims to fosters more trust among stakeholders and regulatory institutions within the power industry [1], [3].

The proposed flow for the above approach has been diagrammatically presented in the Fig. 1.

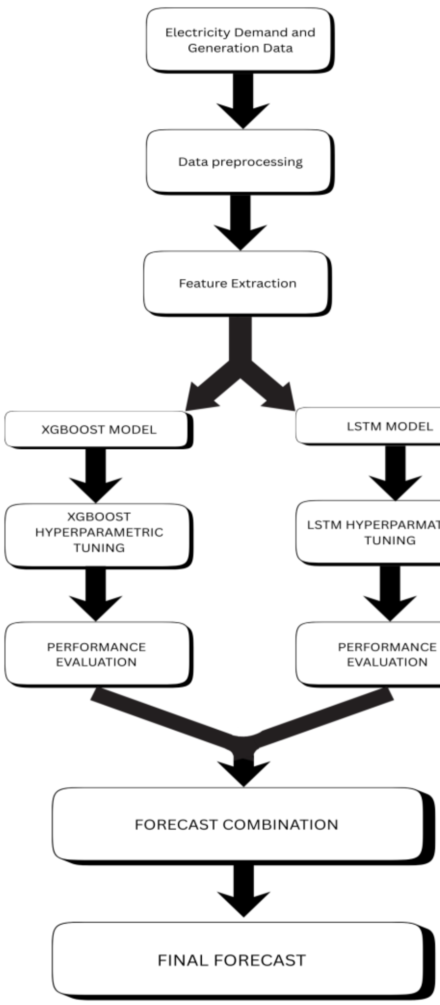

Fig. 1. Proposed framework for electricity demand forecasting using XGBoost-LSTM model.

VI. CHALLENGES AND FUTURE APPROACHES

A. Challenges

Despite the above-mentioned advancements with the adoption of AI based models (ML/DL and hybrid models), there still exist various challenges that hinders their adoption in the practical applications by the power industry and concerned stakeholders :

1. Data Quality & Availability: Many developing countries lack high-resolution, standardized datasets, hindering ML adoption in real-world settings [1], [2], [3].
2. High Complexity and Convergence Issues: AI-based models, including ML/DL based and hybrid models can suffer from high complexity and convergence issues [12]
3. Interpretability: Deep Learning and advanced AI models often act as Black Boxes, meaning their process of Decision making is not clear [1], [3], [5]. The lack of explainability in such models eventually make it difficult for stakeholders, power corporations and regulating institutions to adopt and trust the model's predictions [1], [3], [5].
4. Computational Costs: Deep learning based models are also referred as 'Resource-Hungry' because of their high requirements for more computational resources and long training time, which makes their adoption and implementation less feasible in resource-constrained environments [8], [9], [10], [13].
5. Skill Gaps: The adoption of AI based methodologies presents a dual-edged challenge for the energy sector. On one hand, it exposes the lack of qualified experts and skilled personals present in the industry [2]. On the other hand, it also opens up the gate of new opportunities for the same by bringing more employability in the sector and acting as a catalyst in the upskilling of the present and future workforce across the concerned industries.
6. Cybersecurity & Privacy: With the adoption of smart meters and IoT integration, considering cyber security challenges also becomes critical [2], [8].

B. Future Oppurtunities

• Explainable AI (XAI): Future research should be more centred on improvising and assimilating tools and technologies that provides more interpretability of the decision making process of deep learning models [3]. Making the decision making process more transparent is important for ensuring reliability in the adoption of such AI based forecasting models and this eventually fosters more trust among the stakeholders community and regulatory institutions within the power industry [1], [3].

• Probabilistic Forecasting: Studies indicate that there is an evident shift from the previously adopted deterministic predictions towards the probabilistic intervals [4]. The recent researches are more inclined towards exploring the capability of using hybrid structures and incorporating the highly complex and advanced models such as transformers with deep learning and machine learning models [13]. Probabilistic-Deep-Learning methods are another area of recent developments which integrate deep neural networks with Bayesian inference, offering significant opportunities to quantify uncertainties in forecasting [9].

- Benchmark Datasets: There is a need for the development of community based and shared platforms for fair and transparent comparison of models and datasets [2]. Furthermore contemporary scenario demands development of standardized datasets along with their periodic updates and continuous improvements in order to ensure more reliable comparisons and comprehensive analysis in this field of forecasting [3].

- Integration with Smart Grids: Studies highlights the importance and benefits of adoption of smart grids in the power industry and infrastructure [2]. Accurate electricity demand forecasting is highly improved with the integration of such smart grids and they offer large volume of complex electricity data which is precursor need for the training of AI based models for forecasting electricity load [2], [3].

VII. Conclusion

Electricity demand forecasting is at a crossroads. Statistical approaches remain reliable for linear and long-term predictions but fail in dynamic, nonlinear contexts. ML and DL offer excellent accuracy in short and medium term horizons, especially under seasonal and stochastic variations. Hybrid approaches have emerged as the most promising in the contemporary scenario, combining the interpretability of classical with the predictive power of AI. The future of electricity forecasting lies in explainable, scalable, and renewable-integrated AI systems, ensuring sustainable and resilient grids worldwide.

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